Sphere Five-Point Virasoro Block in the Pillow Frame and the Sphere n-Point Open-Necklace Elliptic h-Recursion

August 29, 2026

Abstract

We write the chiral sphere five-point Virasoro block as a matrix element on the pillow $T^2/\mathbb{Z}_2$. Four external operators sit at the branch points, while the fifth becomes a local operator inserted between two projected propagation segments. We first audit the sphere four-point elliptic recursion directly against a PBW expansion. This fixes the typo in Eq. (E.103) of the string note and separates the geometric c-normalization from the character-absorbed $c-1$ normalization. We then generalize the corrected conformal factor to five points and state the resulting two-channel elliptic h-recursion. The recursion uses the correlated large-weight pillow-character asymptotic as its regular seed. Finally, we extend the construction to a sphere n-point block with $n-4$ local insertions during pillow evolution. The resulting $(n-3)$-edge recursion has a single diagonal large-weight parameter and the same twisted pillow character.

1 Sphere block and pillow geometry

Place the five chiral primaries at

$$V_{d_1}(0), \qquad V_{d_2}(z), \qquad V_{d_3}(1), \qquad V_{d_4}(\infty), \qquad V_{d_5}(t), \tag{1}$$

and choose $(0, z, 1, \infty)$ as the branch quartet. In the comb channel the block is

$$\mathcal{F}_5^{S^2}(z,t) = \langle d_4| V_{d_3}(1)\mathbb{P}_{h_2} V_{d_5}(t)\mathbb{P}_{h_1} V_{d_2}(z)|d_1\rangle. \tag{2}$$

Primary three-point constants are stripped from this definition.

The pillow coordinate and elliptic modulus are

$$w(x) = \frac{1}{\theta_3(\tau)^2}\int_0^x \frac{dy}{\sqrt{y(1-y)(z-y)}}, \qquad \tau = i\frac{K(1-z)}{K(z)}, \tag{3}$$

$$q = e^{i\pi\tau}, \qquad z = \frac{\theta_2(\tau)^4}{\theta_3(\tau)^4}. \tag{4}$$

Here $\theta_j(\tau) = \vartheta_j(0|\tau)$. The fifth puncture maps to

$$w_5 = w(t), \qquad \frac{dw_5}{dt} = \frac{1}{\theta_3(\tau)^2\sqrt{t(1-t)(z-t)}}. \tag{5}$$

The square-root branch in Eqs. (3) and (5) must be continued together with all fractional powers below.

Quantizing the pillow along the A-cycle splits its propagation at w_5. The raw cylinder variables are

$$\widetilde{p}_1 = e^{iw_5}, \qquad \widetilde{p}_2 = e^{i(\pi\tau - w_5)}. \tag{6}$$

After rotating each boundary state by half an A-cycle, define the OPE-aligned local coordinates

$$p_1 = -e^{iw_5}, \qquad p_2 = -e^{i(\pi\tau - w_5)}, \qquad \boxed{p_1 p_2 = q}. \tag{7}$$

They obey

$$\frac{z}{t} = 4p_1 + O(p^2), \qquad t = 4p_2 + O(p^2), \tag{8}$$

which fixes the signs in the recursion variables $4p_i$. The half-period phases are absorbed into the aligned pillow boundary states. The corresponding projected pillow matrix element is

$$\boxed{\mathcal{M}_5(q, w_5) = \langle \psi_{43} | \mathbb{P}_{h_2} p_2^{L_0 - c/24} V_{d_5}^{(w)}(0) p_1^{L_0 - c/24} \mathbb{P}_{h_1} | \psi_{12} \rangle.} \tag{9}$$

The states $|\psi_{12}\rangle$ and $\langle \psi_{43}|$ are prepared by the two lower and two upper corner insertions, respectively. They include the corner regularizations appropriate to the pillow metric.

2 Exact conformal factor

The four branch-point operators produce precisely the Weyl-anomaly and corner factor of the four-point pillow relation. Since the fifth operator is away from the ramification points, it transforms as an ordinary primary,

$$V_{d_5}^{(x)}(t) = \left(\frac{dw_5}{dt}\right)^{d_5} V_{d_5}^{(w)}(w_5). \tag{10}$$

Consequently, the exact five-point generalization of the four-point relation is

$$\mathcal{F}_5^{S^2}(z, t) = \theta_3(\tau)^{c/2 - 4(d_1+d_2+d_3+d_4)} z^{c/24 - d_1 - d_2} (1-z)^{c/24 - d_2 - d_3} \left(\frac{dw_5}{dt}\right)^{d_5} \mathcal{M}_5(q, w_5). \tag{11}$$

The exponent of $1-z$ pairs the operators at z and 1. The $h_3 + h_4$ exponent printed in E.103 of the current string-note PDF is a labeling error, while the $h_2 + h_3$ pairing in E.106 is correct. The four-point audit in Sec. 5 fixes this statement independently of the five-point proposal. It is useful to define, for arbitrary κ,

$$\begin{aligned}\Lambda_5^{(\kappa)}(z, t) := \theta_3(\tau)^{\kappa/2 - 4(d_1+d_2+d_3+d_4) - 2d_5} z^{\kappa/24 - d_1 - d_2} (1-z)^{\kappa/24 - d_2 - d_3} \\ \times [t(1-t)(z-t)]^{-d_5/2}.\end{aligned} \tag{12}$$

Then Eq. (11) becomes

$$\boxed{\mathcal{F}_5^{S^2}(z, t) = \Lambda_5^{(c)}(z, t) \mathcal{M}_5(q, w_5).} \tag{13}$$

3 Cusp normalization and large-weight seed

The factors $4p_1$ and $4p_2$ are fixed by the local degeneration coordinates; they are not introduced by arbitrarily splitting $16q$. We first recall how $16q$ itself arises in the four-point problem. If the external weights are η_i and the exchanged weight is h, the plane OPE normalization gives

$$\mathcal{F}_4^{S^2}(z) \sim z^{h - \eta_1 - \eta_2} \qquad (z \to 0). \tag{14}$$

The corrected pillow relation instead has the leading form

$$\mathcal{F}_4^{S^2}(z) \sim z^{c/24-\eta_1-\eta_2}\mathcal{M}_4(h;q). \tag{15}$$

Matching Eqs. (14) and (15) fixes

$$\mathcal{M}_4(h;q) \sim z^{h-c/24}. \tag{16}$$

Since the modular lambda function obeys

$$z = \frac{\theta_2(\tau)^4}{\theta_3(\tau)^4} = 16q\big(1 - 8q + 44q^2 + O(q^3)\big), \tag{17}$$

we obtain, without taking any large-weight limit,

$$\boxed{\mathcal{M}_4(h;q) \sim (16q)^{h-c/24}.} \tag{18}$$

The cylinder propagator supplies $q^{h-c/24}$; the factor $16^{h-c/24}$ is fixed by the normalization of the pillow boundary states relative to the plane OPE coordinate.

For five points the corresponding argument uses the two independent OPE coordinates

$$x := \frac{z}{t} \longrightarrow 0, \qquad t \longrightarrow 0. \tag{19}$$

The primary term of the plane comb block is

$$\mathcal{F}_5^{S^2}(z,t) \sim z^{h_1-d_1-d_2}t^{h_2-h_1-d_5}. \tag{20}$$

In the same limit $\theta_3 \to 1$, $1-z \to 1$, and Eq. (5) gives

$$\frac{dw_5}{dt} \sim \frac{\epsilon}{t}, \tag{21}$$

where the constant branch phase ϵ is absorbed into the local operator convention. The exact factorization Eq. (13) therefore reduces to

$$\mathcal{F}_5^{S^2}(z,t) \sim z^{c/24-d_1-d_2}t^{-d_5}\mathcal{M}_5. \tag{22}$$

Dividing Eq. (20) by Eq. (22) gives

$$\mathcal{M}_5 \sim z^{h_1-c/24}t^{h_2-h_1} = x^{h_1-c/24}t^{h_2-c/24}. \tag{23}$$

The inverse pillow map fixes the normalization of each local coordinate separately:

$$x = 4p_1\left[1 - 2p_1 + 2p_2 + O(p^2)\right], \qquad t = 4p_2\left[1 + 2p_1 - 2p_2 + O(p^2)\right]. \tag{24}$$

Consequently the primary propagation factor is fixed a priori as

$$\boxed{\mathcal{M}_5 = (4p_1)^{h_1-c/24}(4p_2)^{h_2-c/24}\mathcal{R}_5(h_1,h_2;p_1,p_2),} \qquad \mathcal{R}_5 = 1 + O(p_1,p_2) \tag{25}$$

in the double-cusp expansion. Only after setting $h_1 = h_2 = h$ and using $p_1p_2 = q$ does this imply

$$(4p_1)^{h-c/24}(4p_2)^{h-c/24} = (16q)^{h-c/24}. \tag{26}$$

Thus the five-point factors are derived from the two local coordinates; their product reproduces the four-point normalization as a consistency check, rather than serving as its assumed split.

We now isolate the genuinely additional large-weight proposal. Set

$$h_1 = H, \qquad h_2 = H + a, \tag{27}$$

and take $H \to \infty$ at fixed a, c, d_i, τ, w_5. We assume only that the regular factor in Eq. (25) approaches the pillow character,

$$\mathcal{R}_5(H, H+a; p_1, p_2) \longrightarrow \chi_{\text{pill}}(q), \qquad \chi_{\text{pill}}(q) := \prod_{n=1}^{\infty}(1-q^{2n})^{-1/2}. \tag{28}$$

Equivalently, the complete large-weight asymptotic is

$$\mathcal{M}_5 \sim (4p_1)^{h_1-c/24}(4p_2)^{h_2-c/24}\chi_{\text{pill}}(q). \tag{29}$$

The propagation powers in Eq. (29) are fixed by the double-cusp normalization. Only Eq. (28), the limiting form of the regular factor, is the non-geometric assumption entering the final recursion.

Define the reduced block by

$$\mathcal{M}_5 = (4p_1)^{h_1-c/24}(4p_2)^{h_2-c/24}\chi_{\text{pill}}(q)\mathcal{H}_5(H, a; p_1, p_2), \qquad \mathcal{H}_5 \longrightarrow 1 \quad (H \to \infty). \tag{30}$$

Combining Eqs. (13) and (30) gives the complete sphere block in the geometric c-normalization:

$$\boxed{\mathcal{F}_5^{S^2} = \Lambda_5^{(c)}(z,t)(4p_1)^{h_1-c/24}(4p_2)^{h_2-c/24}\chi_{\text{pill}}(q)\mathcal{H}_5(H, a; p_1, p_2).} \tag{31}$$

The identity

$$\theta_3(\tau)^{1/2}[z(1-z)]^{1/24}(16q)^{-1/24} = \chi_{\text{pill}}(q)^{-1} \tag{32}$$

gives the exactly equivalent Zamolodchikov normalization

$$\boxed{\mathcal{F}_5^{S^2} = \Lambda_5^{(c-1)}(z,t)(4p_1)^{h_1-(c-1)/24}(4p_2)^{h_2-(c-1)/24}\mathcal{H}_5(H, a; p_1, p_2).} \tag{33}$$

Thus χ_{pill} is explicit in Eq. (31), but is already absorbed in Eq. (33); it must not be included a second time in the latter convention.

4 Degenerate data

We use

$$c = 1 + 6Q^2, \qquad Q = b + b^{-1}, \qquad d = \frac{Q^2 - \lambda_d^2}{4}. \tag{34}$$

The degenerate weight and shifted weight are

$$h_{r,s}(c) = \frac{Q^2 - (rb + sb^{-1})^2}{4}, \qquad h_{r,s}(c) + rs = h_{r,-s}(c), \qquad r, s \in \mathbb{Z}_{>0}. \tag{35}$$

In this convention the inverse null-norm slope is

$$A_{r,s}^c = \frac{1}{2} \prod_{\substack{m=1-r,\dots,r \\ n=1-s,\dots,s \\ (m,n)\neq(0,0),(r,s)}} \frac{1}{mb + nb^{-1}}. \tag{36}$$

For weights d_a, d_b, define the fusion polynomial

$$P_{r,s}^c\begin{bmatrix} d_a \\ d_b \end{bmatrix} := \prod_{\substack{p=1-r,\,3-r,\dots,r-1 \\ \ell=1-s,\,3-s,\dots,s-1}} \frac{\lambda_{d_a} + \lambda_{d_b} + pb + \ell b^{-1}}{2} \frac{\lambda_{d_a} - \lambda_{d_b} + pb + \ell b^{-1}}{2}. \tag{37}$$

Equations (36) and (37) fix the normalization of every residue below.

5 Four-point elliptic recursion and normalization audit

Before proposing the five-point recursion, we fix its normalization by a direct four-point calculation. Put four primaries of weights $(\eta_1, \eta_2, \eta_3, \eta_4)$ at $(0, z, 1, \infty)$ and let h propagate in the (12)(34) channel. With three-point constants removed, write

$$\mathcal{F}_4^{S^2}(z) = z^{h-\eta_1-\eta_2} \sum_{N \geq 0} F_N z^N, \qquad F_0 = 1. \tag{38}$$

The PBW coefficients are obtained without elliptic recursion from

$$F_N = \rho_{43,N}^{\mathsf{T}} G_{c,h}^{(N)-1} \rho_{N,12}, \tag{39}$$

where $G_{c,h}^{(N)}$ is the level-N Verma Gram matrix. For example,

$$F_1 = \frac{(h - \eta_1 + \eta_2)(h + \eta_3 - \eta_4)}{2h}. \tag{40}$$

The corrected geometric pillow relation is

$$\boxed{\mathcal{F}_4^{S^2}(z) = \theta_3(\tau)^{c/2-4\sum_{i=1}^4 \eta_i} z^{c/24-\eta_1-\eta_2} (1-z)^{c/24-\eta_2-\eta_3} \mathcal{M}_4(h;q).} \tag{41}$$

Thus the $(1-z)$ exponent pairs the adjacent insertions at z and 1. The $\eta_3 + \eta_4$ pairing in E.103 is a typo; the $\eta_2 + \eta_3$ pairing appearing in E.106 is the correct one.

At large h the pillow matrix element has the regular asymptotic

$$\mathcal{M}_4(h;q) \sim (16q)^{h-c/24} \chi_{\text{pill}}(q), \qquad \chi_{\text{pill}}(q) = \prod_{n \geq 1} (1 - q^{2n})^{-1/2}. \tag{42}$$

Define the reduced elliptic block by

$$\mathcal{M}_4(h;q) = (16q)^{h-c/24} \chi_{\text{pill}}(q) \mathcal{H}_4(h;q). \tag{43}$$

Using Eq. (32) and $\delta = (c-1)/24$, Eqs. (41) and (43) become

$$\boxed{\mathcal{F}_4^{S^2}(z) = (16q)^{h-\delta} z^{\delta-\eta_1-\eta_2} (1-z)^{\delta-\eta_2-\eta_3} \theta_3(\tau)^{(c-1)/2-4\sum_i \eta_i} \mathcal{H}_4(h;q).} \tag{44}$$

The character is explicit in Eq. (43), where the anomaly exponents contain c. It is already absorbed in Eq. (44), where they contain $c-1$. Consequently, retaining the explicit product in E.106 after making all $c \to c-1$ shifts counts the same character twice.

The reduced block obeys the standard elliptic recursion

$$\boxed{\mathcal{H}_4(h;q) = 1 + \sum_{r,s \geq 1} \frac{(16q)^{rs} A_{r,s}^c}{h - h_{r,s}(c)} P_{r,s}^c \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} P_{r,s}^c \begin{bmatrix} \eta_4 \\ \eta_3 \end{bmatrix} \mathcal{H}_4(h_{r,s}(c) + rs; q) \, .} \tag{45}$$

To decide between the printed formulas numerically, we computed Eq. (39) through level six, changed variables with $z = \lambda(q)$, and compared every coefficient with Eq. (45). The three generic test points were

$$(26.215; 0.13, 0.27, 0.41, 0.56; 0.91),$$
$$(31.7; 0.19, 0.34, 0.48, 0.67; 1.07),$$
$$(42.3; 0.16, 0.31, 0.52, 0.73; 1.23),$$

where each row denotes $(c;\eta_1,\eta_2,\eta_3,\eta_4;h)$. The largest relative errors for the corrected normalization were respectively 1.86×10^{-10}, 1.36×10^{-10}, and 7.28×10^{-11}. In contrast, the literal E.103 exponent gives

$$[q]\big(\mathcal{H}_4^{\mathrm{E.103}}-\mathcal{H}_4^{\mathrm{rec}}\big)=16(\eta_2-\eta_4), \tag{46}$$

while retaining the extra character in the $c-1$ formula gives the universal obstruction

$$[q^2]\big(\mathcal{H}_4^{\mathrm{hybrid}}-\mathcal{H}_4^{\mathrm{rec}}\big)=-\frac{1}{2}. \tag{47}$$

The five-point formulas must therefore inherit both lessons: use the adjacent branch-point pairing d_2+d_3, and choose either the geometric c-normalization with χ_{pill} or the $c-1$ normalization without it. In particular, the reduced block used below is equivalently extracted as

$$\boxed{\mathcal{H}_5=\frac{\mathcal{F}_5^{S^2}}{\Lambda_5^{(c-1)}(z,t)(4p_1)^{h_1-(c-1)/24}(4p_2)^{h_2-(c-1)/24}}.} \tag{48}$$

No additional character factor is allowed in the denominator of Eq. (48).

6 Elliptic h-recursion for $\mathcal{H}_5$

Treat $\mathcal{H}_5$ as a meromorphic function of $H=h_1$ while holding $a=h_2-h_1$ fixed. A pole of the first internal edge occurs at $H=h_{r,s}$; replacing the null module by its descendant module sends

$$(H,a)\longmapsto(h_{r,s}+rs,a-rs). \tag{49}$$

A pole of the second edge occurs at $H+a=h_{r,s}$ and sends

$$(H,a)\longmapsto(h_{r,s}-a,a+rs). \tag{50}$$

The resulting two-channel recursion is

$$\boxed{\begin{aligned}\mathcal{H}_5(H,a;p_1,p_2)=1&\\ &+\sum_{r,s\geq1}\frac{(4p_1)^{rs}A^c_{r,s}}{H-h_{r,s}(c)}P^c_{r,s}\begin{bmatrix}d_1\\d_2\end{bmatrix}P^c_{r,s}\begin{bmatrix}h_{r,s}(c)+a\\d_5\end{bmatrix}\mathcal{H}_5(h_{r,s}(c)+rs,a-rs;p_1,p_2)\\ &+\sum_{r,s\geq1}\frac{(4p_2)^{rs}A^c_{r,s}}{H+a-h_{r,s}(c)}P^c_{r,s}\begin{bmatrix}d_4\\d_3\end{bmatrix}P^c_{r,s}\begin{bmatrix}h_{r,s}(c)-a\\d_5\end{bmatrix}\mathcal{H}_5(h_{r,s}(c)-a,a+rs;p_1,p_2)\,.\end{aligned}} \tag{51}$$

All unshifted arguments c,d_i,p_1,p_2 are implicit on the right-hand side. Iterating Eq. (51) automatically generates terms with simultaneous poles in the two internal weights.

When $d_5=0$ and the middle vertex is restricted to the identity channel, $h_2=h_1$ and $p_1p_2=q$. The matrix element then reduces to the four-point pillow propagator, while

$$(4p_1)^{h-c/24}(4p_2)^{h-c/24}=(16q)^{h-c/24}, \tag{52}$$

providing the normalization check against the four-point result.

7 Direct PBW comparison

For an independent check, write the same sphere block in the plane comb frame as

$$\mathcal{F}_5^{\mathrm{PBW}}(z,t) = z^{h_1-d_1-d_2} t^{h_2-h_1-d_5} \sum_{n_1,n_2\geq 0} C_{n_1,n_2} \left(\frac{z}{t}\right)^{n_1} t^{n_2}, \tag{53}$$

where each C_{n_1,n_2} is computed directly by contracting the two PBW Verma bases with inverse Gram matrices and the three descendant three-point tensors. No recursion data enter this calculation.

The inverse covering map in the aligned variables has the form

$$z = 16p_1p_2\, Z(p_1p_2), \qquad t = 4p_2\, Y(p_1,p_2), \qquad \frac{z}{t} = 4p_1 \frac{Z}{Y}, \tag{54}$$

with $Z(0) = Y(0,0) = 1$. Explicitly,

$$Z(q) = \prod_{n\geq 1} \frac{(1+q^{2n})^8}{(1+q^{2n-1})^8}, \tag{55}$$

$$\begin{aligned} Y(p_1,p_2) &= (1+p_1)^2 \prod_{n\geq 1} \frac{(1+q^{2n})^4}{(1+q^{2n-1})^4} \\ &\quad \times \prod_{n\geq 1} \left[\frac{(1+p_1^{2n+1}p_2^{2n})(1+p_1^{2n-1}p_2^{2n})}{(1+p_1^{2n}p_2^{2n-1})(1+p_1^{2n-2}p_2^{2n-1})}\right]^2, \qquad q = p_1p_2. \end{aligned} \tag{56}$$

Substituting Eqs. (54)– (56) into Eq. (53) and dividing by the denominator of Eq. (48) gives a direct PBW series for $\mathcal{H}_5$. Equivalently one may divide by the geometric factor in Eq. (31), including exactly one χ_{pill}. The two procedures agree by Eq. (32).

At first order the direct result is

$$\mathcal{H}_5 = 1 + \frac{2(d_2-d_1)(d_5-a)}{H} p_1 + \frac{2(d_3-d_4)(d_5+a)}{H+a} p_2 + O(p^2), \tag{57}$$

which agrees with the $(r,s) = (1,1)$ terms of Eq. (51). Generic symbolic comparison gives exact agreement through total bidegree three. We then computed the PBW Gram matrices, three-point tensors, and the complete pillow change of variables with exact rational arithmetic through

$$n_1 + n_2 \leq 10, \tag{58}$$

and compared the result with an independent 80-digit numerical evaluation of the h-recursion. This checks all 66 coefficients in each case. We used five fully asymmetric samples:

case	c	(d_1,d_2,d_3,d_4,d_5)	(h_1,h_2)
1	26.215	$(0.17, 0.29, 0.58, 0.71, 0.43)$	$(0.93, 1.08)$
2	31.7	$(0.21, 0.34, 0.63, 0.79, 0.49)$	$(1.03, 1.19)$
3	42.3	$(0.13, 0.31, 0.69, 0.83, 0.47)$	$(0.87, 1.14)$
4	29.3761	$(0.113, 0.367, 1.237, 0.524, 0.811)$	$(1.417, 0.682)$
5	37.219	$(0.907, 0.223, 0.376, 0.658, 1.041)$	$(0.749, 1.533)$

The last two samples deliberately use non-monotone external weights, and case 4 has $a = h_2 - h_1 = -0.735$; hence both signs of the middle-channel weight difference are tested. The discrepancies follow.

case	c	(h_1,h_2)	largest relative discrepancy
1	26.215	$(0.93, 1.08)$	1.363×10^{-74} at $(10,0)$
2	31.7	$(1.03, 1.19)$	1.225×10^{-74} at $(10,0)$
3	42.3	$(0.87, 1.14)$	6.019×10^{-74} at $(10,0)$
4	29.3761	$(1.417, 0.682)$	3.297×10^{-73} at $(10,0)$
5	37.219	$(0.749, 1.533)$	4.539×10^{-73} at $(0,10)$

As a genericity audit, all seven weights are pairwise distinct in each case, all 65 nonconstant PBW coefficients through total order ten are nonzero, and every denominator actually visited by the recursion was recorded. The smallest denominator magnitude over all five cases was 1.232×10^{-3}, far from a pole at 80-digit precision. Thus the comparison comprises $5\times66=330$ coefficients and does not rely on a symmetric choice, a vanishing coefficient, or an accidental null-state specialization. Using the literal h_3+h_4 exponent printed in E.103 instead produces

$$[p_1p_2]\big(\mathcal{H}_5^{\text{literal E.103}}-\mathcal{H}_5^{\text{recursion}}\big)=16(d_2-d_4), \tag{59}$$

which is nonzero for generic weights. The direct PBW calculation therefore both validates Eq. (51) through order ten and fixes the corrected $1-z$ conformal factor. As a second regression test, using the $c-1$ prefactor together with an additional character produces

$$[p_1^2p_2^2]\big(\mathcal{H}_5^{\text{hybrid}}-\mathcal{H}_5^{\text{recursion}}\big)=-\frac{1}{2}, \tag{60}$$

exactly as predicted by the four-point audit. This fixes the character normalization in the five-point computation as well.

8 Comparison with the CCY plane h-recursion

There is a second, already-known h-recursion for the same five-point block: the linear-channel plane recursion of Cho–Collier–Yin (CCY), specialized from their Eq. (3.26). It is important not to identify this representation term by term with Eq. (51). Put

$$x=\frac{z}{t},\qquad H=h_1,\qquad a=h_2-h_1,\qquad e_L=d_1-H,\qquad e_R=d_4-H, \tag{61}$$

and let $G(H,a,e_L,e_R;x,t)$ be the normalized plane descendant series, whose regular term is one. In the notation of this note the CCY recursion is

$$\boxed{\begin{aligned} G(H,a,e_L,e_R;x,t)=1&\\ &+\sum_{r,s\geq1}\frac{x^{rs}A^c_{r,s}}{H-h_{r,s}}P^c_{r,s}\begin{bmatrix}h_{r,s}+e_L\\ d_2\end{bmatrix}P^c_{r,s}\begin{bmatrix}h_{r,s}+a\\ d_5\end{bmatrix}\\ &\qquad\times G(h_{r,s}+rs,a-rs,e_L-rs,e_R-rs;x,t)\\ &+\sum_{r,s\geq1}\frac{t^{rs}A^c_{r,s}}{H+a-h_{r,s}}P^c_{r,s}\begin{bmatrix}h_{r,s}-a\\ d_5\end{bmatrix}P^c_{r,s}\begin{bmatrix}h_{r,s}-a+e_R\\ d_3\end{bmatrix}\\ &\qquad\times G(h_{r,s}-a,a+rs,e_L,e_R;x,t).\end{aligned}} \tag{62}$$

Here and below $h_{r,s}=h_{r,s}(c)$. The pole denominators, the two shifts of (H,a), and the fusion polynomials adjacent to the middle insertion agree exactly with Eq. (51). The boundary fusion polynomials and the expansion variables do not: CCY use (x,t) and send $H,d_1,d_4\to\infty$ while keeping (a, e_L, e_R) fixed. The pillow limit instead keeps every d_i fixed, uses $(4p_1, 4p_2)$, and has the twisted-character asymptotic before the character is stripped. Thus Eq. (51) is not obtained from Eq. (62) by the replacements $x \mapsto 4p_1$ and $t \mapsto 4p_2$ alone; the coordinate change and the complete conformal prefactor must be transformed together.

As a normalization and implementation check, we evaluated Eq. (62) at 80-digit precision and compared all 66 coefficients with $n_1 + n_2 \leq 10$ against the exact rational PBW series at the parameters in Eq. (63) below. The largest relative discrepancy is 1.51×10^{-69}, at coefficient $(n_1, n_2) = (10, 0)$. This independently verifies both the CCY dictionary and the plane-recursion implementation used in the following value test.

9 Value comparison of the two h-recursions with the central-charge recursion

The coefficient audit can be strengthened to a comparison of the block at finite moduli. We choose

$$\begin{aligned} c = 31.7, \qquad (d_1, d_2, d_3, d_4, d_5) &= (0.21, 0.34, 0.63, 0.79, 0.49), \\ (h_1, h_2) &= (1.03, 1.19), \end{aligned} \tag{63}$$

fix $z = 0.08$, and scan $0.12 \leq t \leq 0.72$. At each point, the aligned segment nomes are obtained from the exact product map

$$p_1 p_2 = q(z), \qquad t = 4p_2 Y(p_1, p_2), \tag{64}$$

not from its leading cusp approximation. We evaluate the pillow recursion through total elliptic degree ten and restore the complete plane block with Eq. (33). We also evaluate the CCY plane h-recursion, Eq. (62), through total degree ten in $(z/t, t)$. Independently, the CCY fixed-weight c-recursion gives the same plane descendant series through total degree 22.

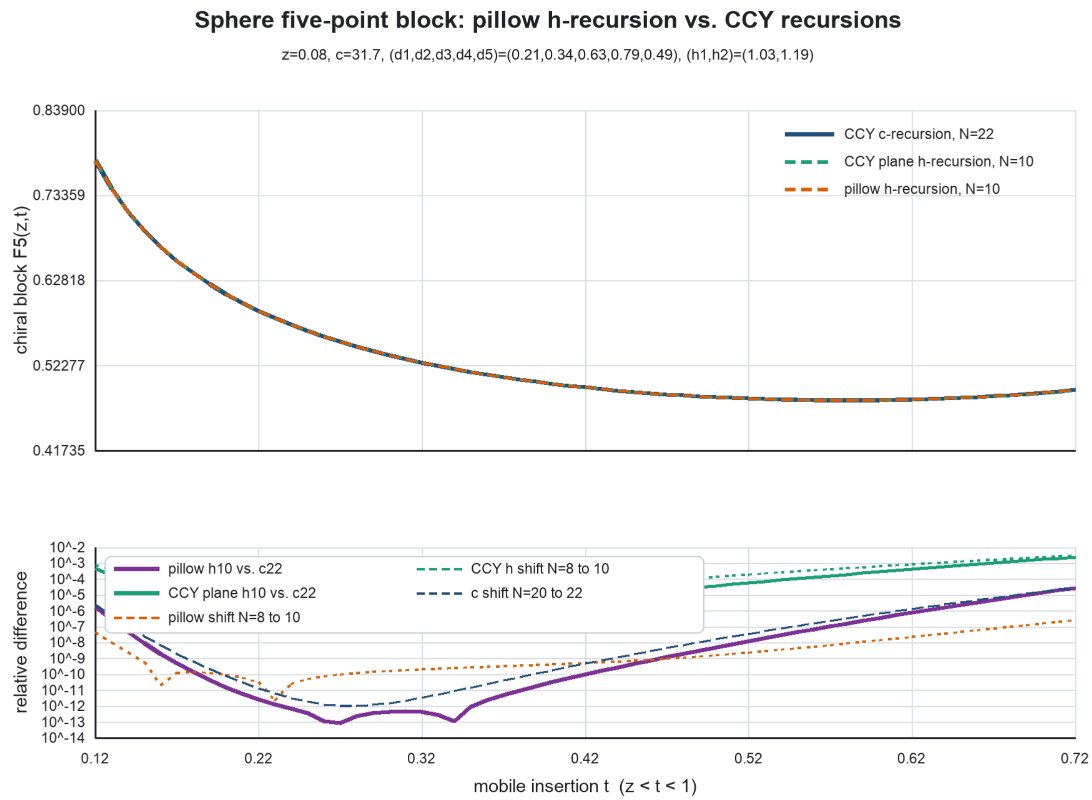

Figure 1: The reconstructed pillow h-recursion, the CCY plane h-recursion, and the independent CCY c-recursion at fixed $z = 0.08$. The lower panel shows the two relative differences from the order-22 c-recursion together with the changes under $8 \to 10$ in each h-recursion and $20 \to 22$ in the c-recursion.

For all 61 sample points, each h–c difference is smaller than the sum of its displayed h-recursion shift and the c-recursion shift. For the pillow representation the median and maximum relative differences are 1.22×10^{-9} and 2.77×10^{-5}; for the CCY plane h-recursion they are 1.81×10^{-5} and 2.52×10^{-3}. At the central point $t = 0.42$, the respective differences are 9.62×10^{-11} and 6.33×10^{-6}. At the least convergent endpoint $t = 0.72$, the order shifts are 2.77×10^{-7} for the pillow h-recursion, 3.20×10^{-3} for the CCY plane h-recursion, and 2.98×10^{-5} for the c-recursion. Therefore both h-recursions converge to the same block within independently measured truncation uncertainty, while the pillow variables give substantially faster convergence at equal total order over this scan.

10 Sphere n-point open-necklace extension

The five-point construction extends without changing the boundary-state geometry. Let

$$m = n - 4, \qquad r = m + 1 = n - 3, \tag{65}$$

where m is the number of ordinary mobile insertions and r is the number of internal propagation segments. Keep the four branch-point weights (d_1, d_2, d_3, d_4) and denote the mobile weights by

$\mu_1, \ldots, \mu_m$. If $w_j = w(t_j)$ are ordered along the chosen pillow evolution contour, set $w_0 = 0$, $w_r = \pi\tau$, and define raw segment parameters

$$\widetilde{p}_i = e^{i(w_i - w_{i-1})}, \qquad \prod_{i=1}^{r} \widetilde{p}_i = q. \tag{66}$$

The phases are fixed by the collision limits of the chosen comb channel. For the nested real ordering

$$0 < z < t_1 < \cdots < t_m < 1, \qquad x_1 = \frac{z}{t_1}, \quad x_i = \frac{t_{i-1}}{t_i} \ (2 \le i \le m), \quad x_r = t_m, \tag{67}$$

the simultaneous comb cusp is $x_i \to 0$. The degenerating pillow map gives

$$e^{iw(t_j)} = -\frac{z}{4t_j}\big[1 + O(\boldsymbol{x})\big], \qquad q = \frac{z}{16}\big[1 + O(z)\big]. \tag{68}$$

It follows directly that, for $r \ge 2$,

$$\begin{aligned} \widetilde{p}_1 &= -\frac{x_1}{4}\big[1 + O(\boldsymbol{x})\big], & \widetilde{p}_i &= x_i\big[1 + O(\boldsymbol{x})\big] & & (2 \le i \le r-1), \\ \widetilde{p}_r &= -\frac{x_r}{4}\big[1 + O(\boldsymbol{x})\big]. \end{aligned} \tag{69}$$

Therefore the OPE-aligned phases are not arbitrary:

$$\boxed{\varepsilon_1 = \varepsilon_r = -1, \qquad \varepsilon_i = +1 \quad (2 \le i \le r-1), \qquad p_i = \varepsilon_i \widetilde{p}_i.} \tag{70}$$

Their product is one, and hence

$$\boxed{\prod_{i=1}^{r} p_i = q.} \tag{71}$$

For another analytic continuation the phases are continued from Eq. (70); they are not independently selectable.

Up to translations that place every middle vertex at its sewing seam, the projected matrix element is

$$\boxed{\mathcal{M}_n = \langle \psi_{43} | \mathbb{P}_{h_r} p_r^D V_{\mu_m} \mathbb{P}_{h_{r-1}} p_{r-1}^D \cdots V_{\mu_1} p_1^D \mathbb{P}_{h_1} | \psi_{12} \rangle, \qquad D = L_0 - \frac{c}{24}.} \tag{72}$$

All primary three-point constants are stripped. Since every mobile operator is inserted away from a branch point, it transforms by its ordinary primary Jacobian. In the present conventions,

$$w(t) = \frac{1}{\theta_3(\tau)^2} \int_0^t \frac{dy}{\sqrt{y(1-y)(z-y)}}, \qquad \frac{dw_j}{dt_j} = \frac{1}{\theta_3(\tau)^2 \sqrt{t_j(1-t_j)(z-t_j)}}. \tag{73}$$

10.1 Exact sphere-to-pillow conformal factor and extraction recipe

The relation needed for an actual computation is exact, before taking any large-internal-weight limit. It is the n-point generalization of the four-point pillow transformation:

$$\boxed{\begin{aligned} \mathcal{F}_n^{S^2}(z; \boldsymbol{t}) = \theta_3(\tau)^{c/2 - 4\sum_{a=1}^4 d_a} z^{c/24 - d_1 - d_2} (1-z)^{c/24 - d_2 - d_3} \\ \times \prod_{j=1}^{m} \left(\frac{dw_j}{dt_j}\right)^{\mu_j} \mathcal{M}_n(\tau; \boldsymbol{w}). \end{aligned}} \tag{74}$$

Thus there are no additional pairwise prime-form factors between the mobile insertions. Such dependence remains inside the matrix element $\mathcal{M}_n$; the coordinate change supplies one local Jacobian per mobile primary. All square-root branches are continued together with the pillow map.

For later use, define for arbitrary κ

$$\begin{aligned}\Lambda_n^{(\kappa)} &:= \theta_3(\tau)^{\kappa/2-4\sum_{a=1}^4 d_a-2\sum_{j=1}^m \mu_j} z^{\kappa/24-d_1-d_2}(1-z)^{\kappa/24-d_2-d_3} \\ &\quad\times\prod_{j=1}^m [t_j(1-t_j)(z-t_j)]^{-\mu_j/2}.\end{aligned} \tag{75}$$

Equations (73) and (74) are then simply

$$\boxed{\mathcal{F}_n^{S^2} = \Lambda_n^{(c)}\mathcal{M}_n.} \tag{76}$$

Once the effective plumbing coordinates ρ_i are fixed below, the two equivalent useful forms are

$$\boxed{\mathcal{F}_n^{S^2} = \Lambda_n^{(c)}\left[\prod_i \rho_i^{h_i-c/24}\right]\chi_{\text{pill}}(q)\mathcal{H}_n} \tag{77}$$

$$\boxed{\mathcal{F}_n^{S^2} = \Lambda_n^{(c-1)}\left[\prod_i \rho_i^{h_i-(c-1)/24}\right]\mathcal{H}_n} \tag{78}$$

The second line is the most convenient PBW-to-elliptic conversion. In particular, the reduced elliptic block is extracted from a direct sphere block by

$$\boxed{\begin{aligned}\mathcal{H}_n(H,\boldsymbol{a};\boldsymbol{p}) &= \frac{\mathcal{F}_n^{S^2}(z;\boldsymbol{t})}{\Lambda_n^{(c-1)}(z;\boldsymbol{t})\prod_{i=1}^r \rho_i^{h_i-(c-1)/24}} \\ &= \frac{\mathcal{F}_n^{S^2}(z;\boldsymbol{t})}{\Lambda_n^{(c)}(z;\boldsymbol{t})\prod_{i=1}^r \rho_i^{h_i-c/24}\chi_{\text{pill}}(q)}.\end{aligned}} \tag{79}$$

The equality follows from

$$\theta_3(\tau)^{1/2}[z(1-z)]^{1/24}(16q)^{-1/24} = \chi_{\text{pill}}(q)^{-1}, \tag{80}$$

together with $\prod_i \rho_i = 16q$. Notice that all mobile-position and mobile-weight factors cancel in passing between the two conventions.

The computational recipe is therefore:

1. choose the branch quartet $(0, z, 1, \infty)$ and order the mobile insertions t_j in the desired open-necklace channel;
2. evaluate $w_j = w(t_j)$ and form the raw segment nomes $\widetilde{p}_i$;
3. determine the phases from the nested OPE collision limit, then form p_i and the endpoint-normalized ρ_i;
4. compute $\Lambda_n^{(c-1)}$ and divide the direct PBW sphere block by either denominator in Eq. (79);
5. re-expand the result in the aligned p_i and apply the edge h-recursion to the resulting $\mathcal{H}_n$.

10.2 Endpoint normalization and the diagonal large-weight seed

The factors of 4 belong to the two regulated pillow boundary states, not to every propagation segment. The effective plumbing parameters are therefore

$$\boxed{\rho_i = 4^{\delta_{i1}+\delta_{ir}} p_i, \qquad \prod_{i=1}^{r} \rho_i = 16q.} \tag{81}$$

Equations (69) and (70) now give the stronger componentwise normalization for $r \geq 2$

$$\boxed{\rho_i = x_i\big[1 + O(\boldsymbol{x})\big], \qquad i = 1, \ldots, r.} \tag{82}$$

For $r = 1$, define $x_1 = z$; then $\rho_1 = 16q = x_1[1+O(x_1)]$ gives the same statement. This formula also covers $r = 1$: both Kronecker deltas contribute and give $\rho_1 = 16q$, the four-point variable. For $r = 2$ it gives $(\rho_1, \rho_2) = (4p_1, 4p_2)$, while for $r > 2$ the intermediate segments carry no additional factor of 4. Assigning $4p_i$ to every segment would instead give $4^r q$ and would fail the identity-insertion reduction to the four-point block.

Now set

$$h_i = H + a_i, \qquad a_1 = 0, \qquad H \longrightarrow \infty, \tag{83}$$

with c, d_a, μ_j, a_i, p_i fixed. In an orthonormal descendant basis the jth middle vertex obeys the same CCY diagonal limit used at five points,

$$\frac{\langle H + a_{j+1}, A|V_{\mu_j}|H + a_j, B\rangle}{\|H + a_{j+1}, A\|\,\|H + a_j, B\|} \longrightarrow \delta_{AB}. \tag{84}$$

Thus all oscillator labels along the open necklace agree. Their descendant propagation is $\prod_i p_i^{|A|} = q^{|A|}$, and the two terminal wavefunctions contract them exactly as in the four-point pillow amplitude. Consequently no new character is generated by an additional middle insertion. Coefficientwise in the segment parameters,

$$\boxed{\mathcal{M}_n \underset{H\to\infty}{\sim} \left[\prod_{i=1}^{r} \rho_i^{h_i - c/24}\right] \chi_{\text{pill}}(q), \qquad \chi_{\text{pill}}(q) = \prod_{k\geq 1}(1-q^{2k})^{-1/2}.} \tag{85}$$

Equivalently,

$$\mathcal{M}_n \sim (16q)^{H-c/24} \left[\prod_{i=2}^{r} \rho_i^{a_i}\right] \chi_{\text{pill}}(q). \tag{86}$$

The product involving a_i is finite primary propagation; the oscillator regular part depends only on q. For example, at six points $r = 3$ and $(\rho_1, \rho_2, \rho_3) = (4p_1, p_2, 4p_3)$, so this finite factor is $p_2^{a_2}(4p_3)^{a_3}$ rather than $(4p_2)^{a_2}(4p_3)^{a_3}$.

Define the reduced block by

$$\mathcal{M}_n = \left[\prod_{i=1}^{r} \rho_i^{h_i - c/24}\right] \chi_{\text{pill}}(q)\, \mathcal{H}_n(H, \boldsymbol{a}; \boldsymbol{p}), \qquad \mathcal{H}_n \longrightarrow 1. \tag{87}$$

Substituting this definition into Eq. (76) reproduces the exact full-block relations (77)–(78).

10.3 The $(n-3)$-edge recursion

At a pole on edge i,

$$H + a_i = h_{\alpha,\beta}(c). \tag{88}$$

The two fusion factors adjacent to that edge are conveniently written

$$\mathcal{L}_{i;\alpha,\beta} := \begin{cases} P^c_{\alpha,\beta}\begin{bmatrix} d_1 \\ d_2 \end{bmatrix}, & i = 1, \\ P^c_{\alpha,\beta}\begin{bmatrix} h_{\alpha,\beta} + a_{i-1} - a_i \\ \mu_{i-1} \end{bmatrix}, & i > 1, \end{cases} \tag{89}$$

$$\mathcal{R}_{i;\alpha,\beta} := \begin{cases} P^c_{\alpha,\beta}\begin{bmatrix} h_{\alpha,\beta} + a_{i+1} - a_i \\ \mu_i \end{bmatrix}, & i < r, \\ P^c_{\alpha,\beta}\begin{bmatrix} d_4 \\ d_3 \end{bmatrix}, & i = r. \end{cases} \tag{90}$$

The null-submodule shift is the map

$$\mathsf{T}_{1;\alpha,\beta}: \quad (H, a_2, \ldots, a_r) \longmapsto (h_{\alpha,\beta} + \alpha\beta, a_2 - \alpha\beta, \ldots, a_r - \alpha\beta), \tag{91}$$

$$\mathsf{T}_{i;\alpha,\beta}: \quad (H, a_i) \longmapsto (h_{\alpha,\beta} - a_i, a_i + \alpha\beta), \qquad i > 1, \tag{92}$$

with all $a_{j\neq i}$ fixed in the second line. These shifts replace the singular module by the module of weight $h_{\alpha,\beta} + \alpha\beta$ while preserving every other physical internal weight at the pole.

The proposed sphere n-point pillow recursion is therefore

$$\boxed{\mathcal{H}_n(H, \boldsymbol{a}; \boldsymbol{p}) = 1 + \sum_{i=1}^{r} \sum_{\alpha,\beta\geq 1} \frac{\rho_i^{\alpha\beta} A^c_{\alpha,\beta} \mathcal{L}_{i;\alpha,\beta} \mathcal{R}_{i;\alpha,\beta}}{H + a_i - h_{\alpha,\beta}(c)} \mathcal{H}_n(\mathsf{T}_{i;\alpha,\beta}(H, \boldsymbol{a}); \boldsymbol{p})\,.} \tag{93}$$

For two segments this formula reproduces Eq. (51) term by term. For one segment it is the four-point elliptic recursion. If all mobile operators are set to the identity and all h_i are identified, the projectors must first be composed; Eq. (81) then reduces the entire propagation to $(16q)^{L_0 - c/24}$. This is a matrix-element reduction and should not be performed term by term after separating coincident pole families.

The new part of this proposal is therefore not a new regular function for each insertion. Once the fixed-external-weight pillow boundary asymptotic is accepted, the CCY diagonal-vertex limit forces every additional mobile operator to act as the identity on the leading oscillator labels. The same single twisted character supplies the regular term for all n.

10.4 Six-point PBW consistency certificate

The first genuine three-edge test is the comb block at $(0, z, t_1, t_2, 1, \infty)$. Put

$$x_1 = \frac{z}{t_1}, \qquad x_2 = \frac{t_1}{t_2}, \qquad x_3 = t_2, \qquad q = p_1 p_2 p_3, \tag{94}$$

and define the two exact mobile-coordinate units by

$$t_1 = 4p_2p_3Y_1, \qquad t_2 = 4p_3Y_2, \qquad z = 16qZ. \tag{95}$$

Here Y_1 is the five-point pillow product with aggregate nomes (p_1, p_2p_3), Y_2 is the same product with aggregate nomes (p_1p_2, p_3), and $Z = z/(16q)$. The direct PBW variables therefore obey the exact identities

$$\boxed{x_1 = 4p_1\frac{Z}{Y_1}, \qquad x_2 = p_2\frac{Y_1}{Y_2}, \qquad x_3 = 4p_3Y_2.} \tag{96}$$

This equation tests the central assertion that the middle plumbing segment has no factor of 4.

The plane primary factor used in the independent PBW computation is

$$z^{h_1-d_1-d_2}t_1^{h_2-h_1-d_3}t_2^{h_3-h_2-d_4}. \tag{97}$$

After multiplying by the inverse of the exact conformal factor in Eq. (74), the remaining unit-series factor is

$$\begin{aligned} Z^{h_1-c/24}Y_1^{a_2}Y_2^{a_3-a_2}[(1-t_1)(1-z/t_1)]^{d_3/2}[(1-t_2)(1-z/t_2)]^{d_4/2} \\ \times\, \theta_3^{-c/2+4(d_1+d_2+d_5+d_6)+2(d_3+d_4)}(1-z)^{-c/24+d_2+d_5}\chi_{\mathrm{pill}}^{-1}. \end{aligned} \tag{98}$$

Together with a factor $4^{n_1+n_3}$ on a PBW monomial $x_1^{n_1}x_2^{n_2}x_3^{n_3}$, this is the complete direct conversion to $\mathcal{H}_6$.

Two independent executable checks were performed. First, all 20 coefficients with $n_1+n_2+n_3 \le 3$ were computed with generic symbolic $(H, a_2, a_3, d_1, \ldots, d_6)$ and $c = 1+6(b+b^{-1})^2$. Sparse polynomial cross-multiplication proves that every PBW coefficient equals the corresponding three-edge recursion coefficient exactly, including all ten coefficients of total degree three.

Second, all 286 coefficients with total degree at most ten were tested at each of ten asymmetric generic internal-weight points. This ensemble includes strictly ascending and descending triples, central peaks and valleys, two near-equal nonmonotone triples, and widely separated weights. The plane PBW Gram matrices, three-point tensors, coordinate transformation, and conformal-factor stripping were evaluated with exact rational arithmetic; the recursion was evaluated independently at 80 decimal digits. Before each PBW evaluation, an exhaustive traversal of every affine recursion shift through total degree ten rejected any point within 10^{-4} of a visited Kac denominator. The choices and audit results were

case	(h_1, h_2, h_3)	minimum pole denominator	maximum relative error
1	$(0.9371, 1.0837, 1.3321)$	1.60×10^{-3}	2.85×10^{-71}
2	$(1.4193, 0.6871, 1.1098)$	1.34×10^{-3}	5.16×10^{-71}
3	$(0.7517, 1.5372, 0.8894)$	9.97×10^{-3}	4.24×10^{-70}
4	$(1.6423, 1.2711, 0.8237)$	5.73×10^{-3}	3.22×10^{-71}
5	$(0.9043, 0.9187, 0.8871)$	2.80×10^{-3}	7.39×10^{-71}
6	$(0.6189, 1.2473, 1.8891)$	4.98×10^{-3}	2.53×10^{-70}
7	$(1.1267, 1.8429, 0.5679)$	1.71×10^{-3}	3.44×10^{-69}
8	$(1.7731, 0.5427, 1.2863)$	4.82×10^{-3}	7.38×10^{-73}
9	$(1.1047, 1.1299, 1.0873)$	6.63×10^{-3}	5.43×10^{-72}
10	$(0.4831, 1.9637, 1.2249)$	6.00×10^{-4}	2.29×10^{-69}

Cases $1, 4, 5, 10$ use $c = 26.215$, cases $2, 6, 7$ use $c = 29.3761$, and cases $3, 8, 9$ use $c = 37.219$. Each group reuses one generic external-weight tuple; the exact tuples are stored in the executable certificate. All nine weights are distinct at every point, no nonconstant PBW coefficient vanishes, and no visited recursion denominator is smaller than the values shown. Thus 2860 numerical coefficients pass. The global worst case is coefficient $(0, 10, 0)$ in case 7, precisely a pure excitation of the unrescaled middle segment.

The executable certificates are the symbolic-order-three and numerical-order-ten six-point checkers in `Code/sphere_five_kummer_h_recursion/`.

10.5 Six-point value comparison with the central-charge recursion

We next compare values of the reconstructed pillow block with the independent fixed-internal-weight c-recursion. Keep the last two mobile insertions fixed and vary the first one in the ordered real cell:

$$t_1 = 0.32, \qquad t_2 = 0.62, \qquad 0.015 \le z \le 0.20. \tag{99}$$

At every z, set $q = q(z)$ and let $T(p, q)$ denote the exact five-point product $4pY(q/p, p)$. The six-point inverse map is particularly simple:

$$\boxed{T(P_{23}, q) = t_1, \qquad T(p_3, q) = t_2, \qquad p_1 = \frac{q}{P_{23}}, \qquad p_2 = \frac{P_{23}}{p_3}.} \tag{100}$$

Thus $P_{23} = p_2 p_3$ and $p_1 p_2 p_3 = q$ hold exactly; no multivariate root search is required. We solved the two monotone real equations using the infinite product itself. The largest coordinate residual in the scan was 7.8×10^{-67}.

The generic parameters were

$$\begin{gathered} c = 26.215, \qquad (d_1, \ldots, d_6) = (0.17, 0.29, 0.43, 0.58, 0.71, 0.86), \\ (h_1, h_2, h_3) = (0.9371, 1.0837, 1.3321), \end{gathered} \tag{101}$$

where the external weights are ordered at $(0, z, t_1, t_2, 1, \infty)$. With $\delta = (c-1)/24$, the pillow computation was restored to the plane block as

$$\mathcal{F}_6^{(h,N)} = \Lambda_6^{(c-1)} (4p_1)^{h_1-\delta} p_2^{h_2-\delta} (4p_3)^{h_3-\delta} \mathcal{H}_6^{(N)}(p_1, p_2, p_3). \tag{102}$$

This is compared with

$$\mathcal{F}_6^{(c,N)} = z^{h_1-d_1-d_2} t_1^{h_2-h_1-d_3} t_2^{h_3-h_2-d_4} \sum_{n_1+n_2+n_3 \le N} C^{(c)}_{n_1 n_2 n_3} \left(\frac{z}{t_1}\right)^{n_1} \left(\frac{t_1}{t_2}\right)^{n_2} t_2^{n_3}, \tag{103}$$

where $C^{(c)}$ is generated by the fixed-weight central-charge recursion. The displayed curves use $N = 10$ for the elliptic recursion and $N = 20$ for the c-recursion. Their observed truncation scales are measured by $8 \to 10$ and $18 \to 20$, respectively.

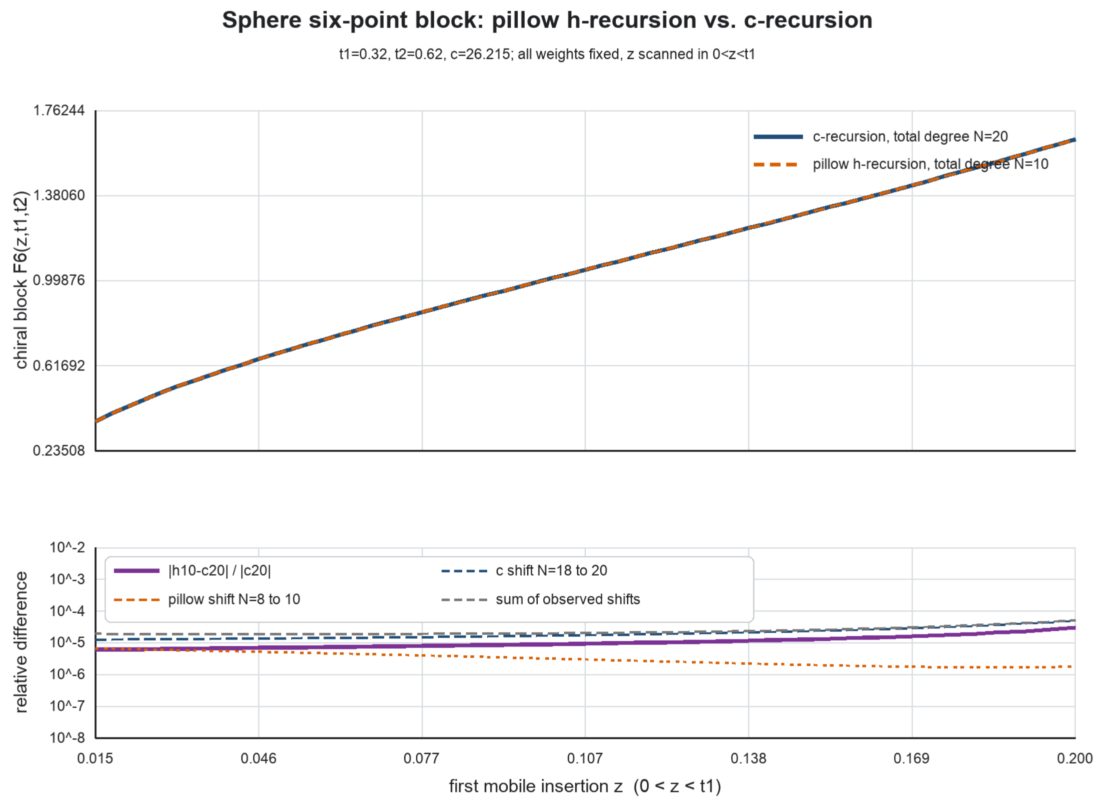

Figure 2: The sphere six-point chiral block as the first mobile insertion z varies while t_1, t_2 and all weights remain fixed. The upper-panel curves are visually indistinguishable. The lower panel compares their relative difference with the independently observed elliptic and central-charge truncation shifts.

Across all 61 points, the median relative difference is 9.25×10^{-6} and the maximum is 2.91×10^{-5}, attained at $z = 0.20$. At that endpoint the elliptic $8 \to 10$ shift is 1.78×10^{-6} and the c-recursion $18 \to 20$ shift is 4.98×10^{-5}. Point by point, the disagreement is smaller than the sum of these two shifts; its maximum ratio to that sum is 0.563. The six-point elliptic and central-charge recursions therefore agree throughout the scan within an uncertainty estimated independently from each series.

Posterior convergence of the order-ten elliptic truncation. The $8 \to 10$ shift used in Fig. 2 is deliberately conservative: it includes two complete shells already retained in the order-ten answer. To estimate the omitted tail more directly, we extended only the elliptic recursion to total degree twelve. For

$$\epsilon_N(z) := \frac{|\mathcal{F}_6^{(h,N)}(z) - \mathcal{F}_6^{(h,N-1)}(z)|}{|\mathcal{F}_6^{(h,N)}(z)|}, \tag{104}$$

the shell sizes over the same 61 points are

shell	median ϵ_N	maximum ϵ_N
$N = 10$	4.91×10^{-7}	1.54×10^{-6}
$N = 11$	2.26×10^{-7}	6.47×10^{-7}
$N = 12$	1.07×10^{-7}	2.62×10^{-7}

There is mild pointwise alternation, so the most useful diagnostic is the combined two-shell change

$$\eta_{10}^{(12)}(z) := \frac{|\mathcal{F}_6^{(h,12)}(z) - \mathcal{F}_6^{(h,10)}(z)|}{|\mathcal{F}_6^{(h,12)}(z)|}. \tag{105}$$

Its minimum, median, and maximum are respectively 1.18×10^{-7}, 3.19×10^{-7}, and 9.09×10^{-7}. The maximum occurs at $z = 0.015$; at the midpoint $z = 0.1075$ and the opposite endpoint $z = 0.20$ the values are 3.19×10^{-7} and 1.18×10^{-7}. Thus a conservative posterior statement for this real-moduli scan is

$$\boxed{\text{relative truncation uncertainty of the order-ten elliptic block is below } 10^{-6}.} \tag{106}$$

This is an empirical next-shell estimate, not a rigorous tail bound. It is nevertheless much smaller than the maximum 2.90×10^{-5} elliptic–c difference and the maximum 4.98×10^{-5} c-recursion $18 \to 20$ shift. Hence the value comparison is limited by the slower central-charge series, not by the order-ten elliptic truncation.

The script, CSV table, JSON summary, and vector/raster plots are stored in `Code/sphere_five_kummer_h_recursion/` and `DataSet/h-Recursion/`.

The formulas above are chiral. The full nonchiral block is obtained by multiplying by the antiholomorphic counterpart with barred weights and coordinates.